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BHARATIYA ANTARIKSH HACKATHON 2026



Team Name : Syntaxz squad

Team Leader Name : DEEPAK S

Problem Statement : AI-Driven Automated Crop Type, Moisture Stress Detection & Irrigation Advisory

Team Members

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Team Member-1:

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Opportunity & USP

THE PROBLEM

Conventional crop monitoring is manual, slow & expensive

Cloud cover limits optical satellite utility for 4–6 months

No integrated system for crop type + stress + irrigation advice

Farmers lack timely, actionable data to optimize irrigation

Existing models fail across varying phenological stages

OUR UNIQUE APPROACH

Multi-source fusion: Optical (LISS-IV/III, AWiFS, Sentinel-2) + SAR (Sentinel-1, EOS-4, NISAR)

All-weather monitoring — SAR penetrates cloud cover

Stage-aware models: phenology-specific ML inference per growth stage

End-to-end AI pipeline: Classification → Stress Detection → Irrigation Advisory

Indigenous satellite data (IRS, EOS) prioritized for Indian farmlands

Scalable to district/state level via GEE / ISRO Bhuvan cloud

Key Features of the Solution



Multi-Sensor Data Fusion

Seamlessly integrates LISS-IV, AWiFS, Sentinel-2 optical data with Sentinel-1 & EOS-4 SAR to overcome individual sensor limitations. NISAR-ready pipeline.



AI Crop Type Classification

Random Forest, XGBoost & CNN-based classifiers distinguish 15+ crop types (rice, wheat, cotton, sugarcane etc.) with >90% OA using seasonal time-series stacks.



Phenological Stage Mapping

Automated detection of sowing, vegetative, flowering, grain-fill & harvest stages using temporal NDVI/SAR coherence signatures per crop.



Moisture Stress Detection

CWSI, NDWI, LSWI & backscatter anomaly indicators fused via ensemble model to detect early, mild and severe moisture stress classes.



Irrigation Advisory Engine

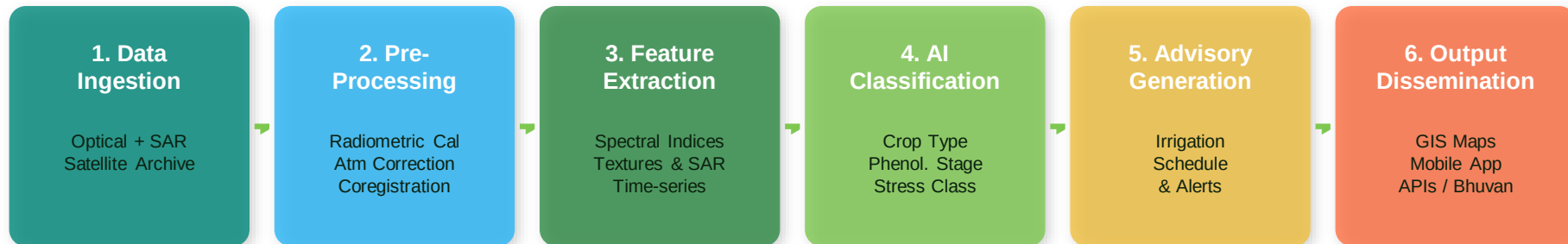
Rule-based + ML hybrid advisory generating field-level irrigation schedules with quantity estimates (mm/ha), timing and crop-stage context.



Scalable Cloud Processing

Implemented on Google Earth Engine + ISRO Bhuvan cloud infrastructure for district-to-national scale deployment with near real-time outputs.

System Process Flow



✦ OPTICAL PATH: LISS-IV → LISS-III → AWiFS → Sentinel-2 → Landsat-8 → MODIS | Band composites, NDVI, EVI, LSWI, NDWI time-series

✦ SAR PATH: Sentinel-1 C-band → EOS-4 (RISAT-1A) → NISAR (L+S) | Backscatter σ^0 , InSAR coherence, Polarimetric decomposition

AI INFERENCE ENGINE — Stage-wise Model Selection

🌱 Sowing
(SAR backscatter
+ NDVI threshold)

🌿 Vegetative
(NDVI, EVI,
Red-Edge indices)

🌸 Flowering
(Chlorophyll
indices + SAR)

🌾 Grain Fill
(LAI, NDWI,
backscatter ratio)

🌾 Harvest
(NDVI drop +
coherence rise)

OUTPUT → Crop Type Map | Stress Severity Map | Irrigation Advisory (mm/ha/day) | Alert Notifications | API / Dashboard

Wireframes — Dashboard & Output Mocks

FARMER DASHBOARD (Web / Mobile)

[Satellite Crop Map — Color-coded by crop type]
 Rice ■ Wheat ■ Cotton ■ Fallow ■ Stress Zones ●

Crop Type
Rice (Kharif)

Growth Stage
Vegetative (45 DAP)

Stress Level
MILD ▲□

Irrigation Req.
38 mm / ha

Advisory: Irrigate in next 48 hrs | Apply 38 mm | Target: Root zone saturation

STRESS DETECTION OUTPUT — SAMPLE REPORT



No Stress

Very Mild

Mild

Moderate

Severe

Moisture Stress Classification Map | 30m Resolution | Date: Current Season

Index	Value	Status
NDWI	0.12	▲ Low
CWSI	0.68	▲ Stress
LSWI	0.09	✗ Deficit
SAR σ^0	-14 dB	✓ Normal

Confidence Score: 87% | Model: Ensemble (RF + XGBoost + LSTM) | Source: Sentinel-1 + Sentinel-2

System Architecture

LAYER 1: DATA ACQUISITION

LISS-IV / LISS-III / AWiFS (ISRO) · Sentinel-2 (ESA) / Landsat (USGS) · Sentinel-1 SAR / EOS-4 / NISAR · MODIS for coarse-scale temporal



LAYER 2: PRE-PROCESSING & FEATURE ENGINEERING

Radiometric Cal, Sen2Cor, SNAP · Band Math: NDVI, EVI, LSWI, NDWI, SAR σ^0 · Phenological time-series stacks · Cloud masking & SAR speckle filter



LAYER 3: AI / ML MODELS

Crop Classification: RF + XGBoost + CNN + Transformer · Phenology Detection: LSTM time-series model · Moisture Stress: Ensemble (SVM + Gradient Boost + Rule-base) · Irrigation Advisory: Hybrid ML + Expert rules



LAYER 4: CLOUD INFRASTRUCTURE

Google Earth Engine (GEE) for processing at scale · ISRO Bhuvan cloud for archival & dissemination · REST API endpoints for mobile/web integration · Auto-scheduling: daily/weekly updates



LAYER 5: OUTPUTS & DISSEMINATION

GeoTIFF maps, Shapefiles (district/block) · Farmer advisory: SMS / App / Web dashboard · State Dept. integration & PMFBY support · Real-time alerts for critical stress events

Technologies & Tools

Satellite Data Platforms

- ISRO Bhuvan / NRSC Data Archive
- Copernicus Open Access Hub (ESA)
- Google Earth Engine (GEE)
- AWS Open Data Registry (Landsat)
- USGS Earth Explorer

Remote Sensing Tools

- SNAP (ESA) — SAR processing
- Sen2Cor — Sentinel-2 atm correction
- GDAL / Rasterio / Fiona (Python)
- QGIS / ArcGIS Pro
- Google Earth Engine JS/Python API

AI / ML Frameworks

- scikit-learn (RF, SVM, XGBoost)
- TensorFlow / PyTorch (CNN, LSTM)
- Hugging Face Transformers (ViT)
- LightGBM for tabular data
- ONNX for model export/deployment

Cloud & Infrastructure

- Google Earth Engine — compute at scale
- ISRO Bhuvan Cloud (NIC co-hosted)
- Docker + Kubernetes (model serving)
- FastAPI / Flask (REST APIs)
- PostgreSQL + PostGIS (geo database)

Data & Visualization

- NumPy / Pandas / Xarray
- Matplotlib / Seaborn / Plotly
- Folium / Leaflet.js (web maps)
- React.js (farmer dashboard)
- Streamlit (quick prototypes)

Spectral Indices Used

- NDVI, EVI, SAVI, MSAVI
- NDWI, LSWI, MNDWI
- Red-Edge CI, CIre
- SAR VV/VH Backscatter σ^0
- CWSI (Crop Water Stress Index)

Estimated Implementation Cost & Timeline

Component	Details	Est. Cost (INR)	Duration
Data Procurement	ISRO/NRSC archive + ESA Copernicus (free tier)	0 – 2,00,000	Ongoing
Cloud Compute (GEE)	GEE Research Account + ISRO Bhuvan cloud	50,000 – 2,00,000 / yr	Phase 1–3
AI Model Dev & Training	GPU servers (A100) or AWS/GCP spot instances	3,00,000 – 8,00,000	Phase 1–2
Software & Licenses	Open-source stack (SNAP, QGIS, Python, PyTorch)	~0 – 50,000	One-time
Farmer Dashboard Dev	React + Leaflet + FastAPI + Mobile App (Flutter)	4,00,000 – 7,00,000	Phase 2
Field Validation	Ground-truth campaign, sensor deployment	2,00,000 – 5,00,000	Phase 2
Human Resources (2 yr)	RS specialist + AI engineer + domain expert	18,00,000 – 30,00,000	Phase 1–3
TOTAL (Pilot — 2 States)	—	~28–55 Lakh INR	24 months

Phase 1 (M1–8): Data pipeline + AI model dev + GEE integration

Phase 2 (M9–16): Field validation + Dashboard + Pilot in 2 districts

Phase 3 (M17–24): State rollout + ISRO integration

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THANK YOU